

LMS & E-Commerce Super-API Documentation

Comprehensive REST API specification with interactive Swagger integration, modular catalog, cart state engine, Razorpay checkout, and full order fulfillment.

Version: 1.0.0 **Base URL (Prod):** https://mockapi-mauve.vercel.app **Base URL (Dev):** http://localhost:8000

Swagger UI: /api-docs **OpenAPI JSON:** /api-docs.json

 End-to-End Security

Secure dual-mode authentication via JWT Bearer Authorization headers and secure HTTP-Only cookies with role-based access control (User & Admin).

 Razorpay Payment Integration

Complete payment lifecycle with client key discovery, automated order ID creation, cryptographic HMAC-SHA256 signature verification, and webhooks.

 Modern E-Commerce Engine

Hierarchical categories, brand indexing, multi-criteria product search, address book, real-time cart state, wishlist toggles, and promo coupons.

 Learning Management Core

Course creation, video curriculum streaming, lesson Q&A, student reviews, dynamic landing page banners, and sales analytics.

1. Authentication & Users (/api/v1/auth)

Method	Endpoint	Access	Description
POST	/register	Public	Register account & send 4-digit activation code
POST	/activate-user	Public	Verify activation code & token to activate account
POST	/login	Public	Login with credentials; receive tokens & cookies
GET	/logout	Auth	Clear authentication tokens and destroy session

Method	Endpoint	Access	Description
GET	/refresh	Public	Refresh expired access token using refresh token
GET	/me	Auth	Fetch currently authenticated user profile
PUT	/update-user-info	Auth	Update user profile details (name, email)
PUT	/update-user-password	Auth	Change existing user password
GET	/get-users	Admin	Retrieve all registered users
PUT	/update-user-role	Admin	Promote or demote user account role

2. Categories & Brands (/api/v1/category)

Method	Endpoint	Access	Description
GET	/all	Public	Get all active categories with optional parent/child nesting
GET	/single/:idOrSlug	Public	Get single category by ObjectId or URL slug
GET	/brands/all	Public	Get all active brands for filtering and showcases
POST	/create	Admin	Create a new category with icon/image and display order
PUT	/update/:id	Admin	Modify category title, description, or parent relationship
DELETE	/delete/:id	Admin	Permanently delete a category
POST	/brand/create	Admin	Register a new product brand

3. Products Catalog (/api/v1/product)

Method	Endpoint	Access	Description
GET	/all	Public	Search by keyword, filter by category/brand/price, sort, and paginate
GET	/featured	Public	Get featured products for homepage hero showcase
GET	/single/:idOrSlug	Public	Get full product details by ObjectId or slug
GET	/related/:id	Public	Get recommended products in same category
POST	/create	Admin	Create product with price, originalPrice, stock, SKU, and tags
PUT	/update/:id	Admin	Update product specifications, inventory, and media
DELETE	/delete/:id	Admin	Remove product from catalog

4. Address Book (/api/v1/address)

Method	Endpoint	Access	Description
POST	/add	Auth	Add shipping/billing address to user address book
GET	/my-addresses	Auth	Retrieve all saved addresses for active user
PUT	/update/:id	Auth	Update existing address details
DELETE	/delete/:id	Auth	Remove address from address book
PUT	/set-default/:id	Auth	Set address as default destination for quick checkout

5. Shopping Cart & Wishlist

Method	Endpoint	Access	Description
GET	/api/v1/cart	Auth	Get user cart with calculated subtotal, discounts, tax, and total
POST	/api/v1/cart/add	Auth	Add product to cart or increment quantity
PUT	/api/v1/cart/update-quantity	Auth	Adjust quantity of specific item in cart
DELETE	/api/v1/cart/item/:itemId	Auth	Delete item from shopping cart
DELETE	/api/v1/cart/clear	Auth	Empty all items from cart
POST	/api/v1/cart/merge	Auth	Merge guest localStorage cart items upon user authentication
GET	/api/v1/wishlist	Auth	Retrieve user wishlist items
POST	/api/v1/wishlist/toggle	Auth	Toggle product saved state in wishlist
POST	/api/v1/wishlist/move-to-cart/:productId	Auth	Transfer product directly from wishlist into active cart

6. Coupons & Discounts (/api/v1/coupon)

Method	Endpoint	Access	Description
POST	/apply	Auth	Validate and apply promo code with automatic cart price recalculation
POST	/remove	Auth	Remove applied promo coupon from cart
POST	/create	Admin	Create percentage/fixed coupon with min spend, cap, and expiry
GET	/all	Admin	List all promo coupons and redemption counts
DELETE	/delete/:id	Admin	Revoke coupon code

7. Payments & Razorpay (/api/v1/payment)

Method	Endpoint	Access	Description
GET	/razorpay-key	Public	Retrieve client Razorpay Key ID for frontend Checkout SDK
POST	/razorpay-order	Auth	Generate official Razorpay Order ID for cart amount
POST	/verify	Auth	Cryptographically verify HMAC-SHA256 signature after payment
POST	/webhook	Public	Server-to-server webhook endpoint for async payment notifications

8. Orders & Fulfillment (/api/v1/ecommerce/order)

Method	Endpoint	Access	Description
POST	/create	Auth	Place new order (Cash on Delivery or verified online payment)
GET	/my-orders	Auth	View customer order history and real-time delivery status
GET	/single/:id	Auth	Get full order breakdown with line items and delivery tracking
PUT	/cancel/:id	Auth	Cancel order before fulfillment dispatch
GET	/admin/all	Admin	Master order dashboard with status filters and pagination
PUT	/admin/status/:id	Admin	Update status: Pending → Processing → Shipped → Delivered

9. Product Reviews & Ratings (/api/v1/product-reviews)

Method	Endpoint	Access	Description
GET	/product/:productId	Public	Get reviews, star rating distribution, and review count
POST	/add	Auth	Post verified customer review (1-5 stars, title, comment)
PUT	/helpful/:id	Auth	Upvote helpfulness of a review
DELETE	/delete/:id	Auth	Remove review (author or admin)

10. 10-Day Sprint Implementation Roadmap

Step-by-step execution plan for full frontend and backend synchronization:

Day 1: Setup, Database Indexes & Data Seeding

- Verify environment configuration (MongoDB Atlas, Cloudinary, Razorpay keys).
- Deploy composite indexes for catalog filtering and performance tuning.
- Smoke test live Swagger UI at `/api-docs`.

Day 2: Authentication, Email Activation & User Profiles

- Connect registration with 4-digit Ethereum test activation codes.
- Implement automated refresh token rotation and secure HTTP-Only cookies.
- Build User Profile dashboard with avatar upload and password changes.

Day 3: Categories, Brands & Product Catalog

- Build multi-level Category navigation and Brand showcases.
- Implement Product Detail Page (PDP) with image gallery, specs, and price tags.
- Validate Admin CRUD workflows for catalog management.

Day 4: Live Search, Multi-Facet Filters & Sorting

- Implement debounced live keyword search.
- Build multi-filter panel (Categories, Brands, Price Slider, Star Rating).
- Support sorting (price low/high, newest, popular) and responsive pagination.

Day 5: Customer Address Book & Wishlist System

- Implement Address Book (Add, Edit, Delete, Set Default).
- Build Wishlist toggling with persistent user state.
- Implement one-click "Move Wishlist Item to Cart" action.

Day 6: Shopping Cart State Machine & Guest Cart Sync

- Implement guest session localStorage cart and sync on login via `/api/v1/cart/merge`.
- Real-time quantity modification with stock boundary checks.
- Auto-calculate subtotal, discounts, tax, and total pricing.

Day 7: Promotional Coupons & Discount Rules

- Implement percentage and fixed amount coupon validation engine.
- Enforce minimum purchase amount, maximum caps, and expiration limits.
- Build Admin coupon creator and analytics dashboard.

Day 8: Razorpay Gateway Checkout & Signature Verification

- Integrate Razorpay Client SDK using dynamic key from `/api/v1/payment/razorpay-key`.
- Server-side Razorpay Order generation (`/api/v1/payment/razorpay-order`).
- Cryptographic HMAC-SHA256 signature verification and webhook listeners.

Day 9: Order Management, Real-Time Tracking & Fulfillment

- Place orders with Cash on Delivery (COD) or Online Paid options.
- Customer Order History dashboard with real-time status tracker.
- Admin fulfillment console for status updates (Processing → Shipped → Delivered).

Day 10: Product Reviews, End-to-End QA & Production Deployment

- Product reviews submission with 1-5 star ratings and helpfulness upvotes.
- Execute automated test runner across all 44 endpoints.
- Production build verification and Vercel serverless deployment.